

Modeling Wildfire Spread Using Spatial SIR Models and Satellite Data

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Introduction & Motivation

Wildfires are complex spatial phenomena strongly influenced by terrain structure, vegetation density, and environmental conditions. Classical epidemic models, such as the SIR framework, provide a natural basis for describing propagation processes, but they do not account for spatial heterogeneity or localized interactions.

In this project, we develop a stochastic spatial extension of the SIR model using real satellite imagery. The landscape is represented as a spatial lattice where each cell has different vegetation characteristics and flammability.

The objective is to study how terrain heterogeneity and environmental forcing influence wildfire propagation, spread velocity, and total burned area, with particular attention to wind, temperature, humidity, and ignition location.

Mathematical Model

The landscape is discretized into a spatial lattice where each cell (i, j) satisfies

$$X_{i,j}(t) \in \{0, 1, 2\}$$

$0 \rightarrow$ Susceptible $1 \rightarrow$ Burning $2 \rightarrow$ Burned

The ignition probability is defined as

$$P_{i,j}(t) = 1 - \exp(-k\beta_{i,j}N_{i,j}(t))$$

where

$$N_{i,j}(t) = \sum_{(k,l) \in \mathcal{N}_{i,j}} \mathbf{1}_{\{X_{k,l}(t)=1\}}$$

represents the number of burning neighboring cells.

The discrete stochastic dynamics are

$$\begin{aligned} S_{i,j}(t+1) &= S_{i,j}(t) - P_{i,j}(t)S_{i,j}(t) \\ B_{i,j}(t+1) &= B_{i,j}(t) + P_{i,j}(t)S_{i,j}(t) - \gamma_{i,j}B_{i,j}(t) \\ R_{i,j}(t+1) &= R_{i,j}(t) + \gamma_{i,j}B_{i,j}(t) \end{aligned}$$

with spatial burnout rate

$$\gamma_{i,j} = \gamma_0(1 - \beta_{i,j})$$

and minimum burn duration

$$\tau_{i,j} \geq T_{\min}$$

To model wind-driven propagation, the isotropic neighborhood interaction is replaced by

$$N_{i,j}^{\text{eff}} = \sum_{(m,n) \in \mathcal{N}_{i,j}} w_{m,n} \mathbf{1}_{\{X_{m,n}=1\}}$$

leading to the modified ignition probability

$$P_{i,j}(t) = 1 - \exp(-k\beta_{i,j}N_{i,j}^{\text{eff}})$$

Data and Terrain Processing

Real satellite imagery is used to introduce spatial heterogeneity into the model. Vegetation density is extracted from the RGB channels using the index

$$V(x, y) = 2G(x, y) - R(x, y) - B(x, y)$$

where higher values correspond to denser vegetation.

The vegetation index is normalized to obtain a spatial flammability coefficient

$$\beta_{i,j} \in (0, 1]$$

which represents the local susceptibility to fire propagation.

The processed terrain is then resized into a discrete spatial lattice used by the simulation. Each cell (i, j) therefore contains:

$$(X_{i,j}(t), \beta_{i,j})$$

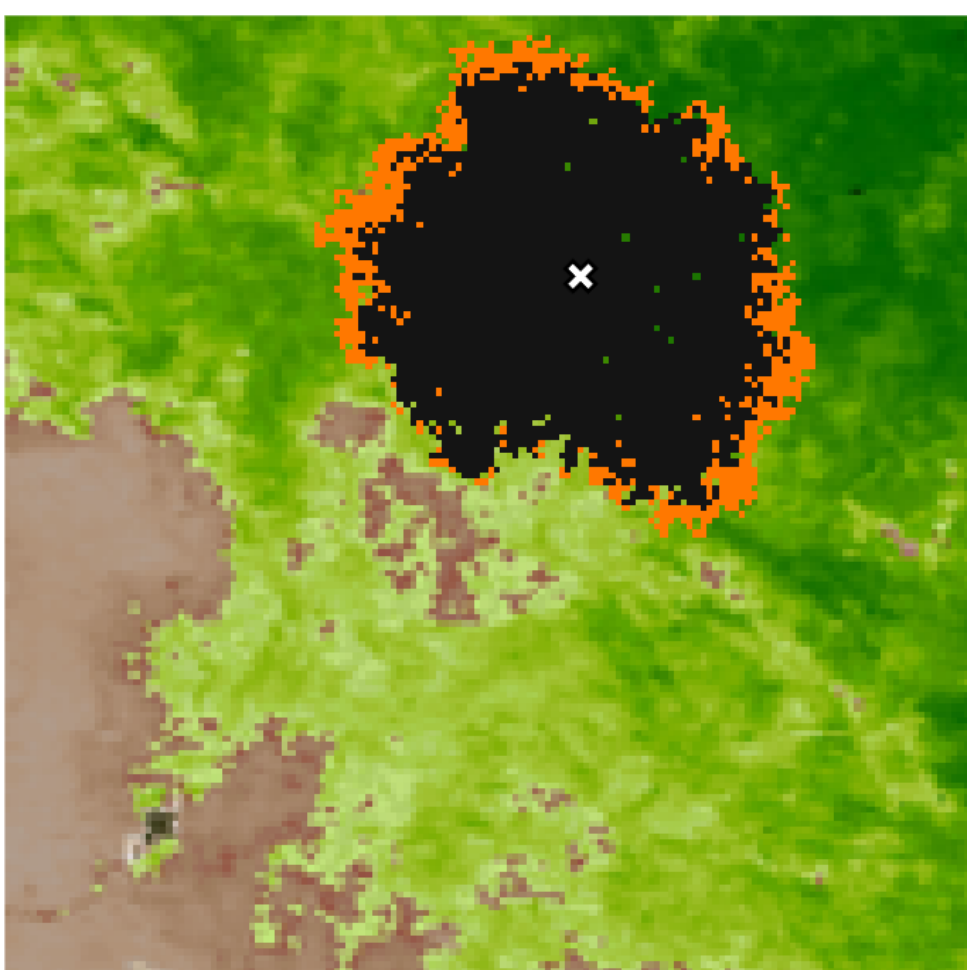
where $X_{i,j}(t)$ denotes the wildfire state and $\beta_{i,j}$ the local flammability.

The resulting maps preserve terrain heterogeneity across California, Australia, and Zamora.

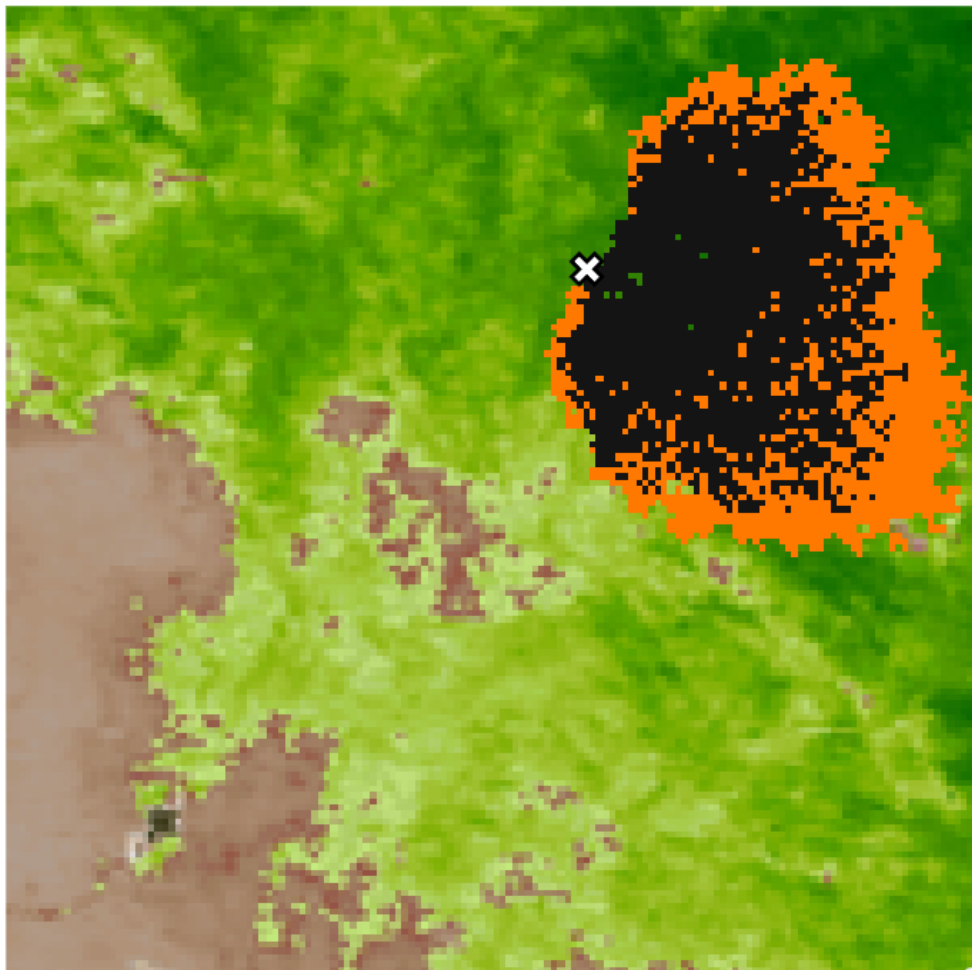
Code and Implementation

The model is implemented numerically using vectorized lattice operations (NumPy roll) and an interactive Dash dashboard for real-time exploration of wildfire propagation under different environmental conditions.

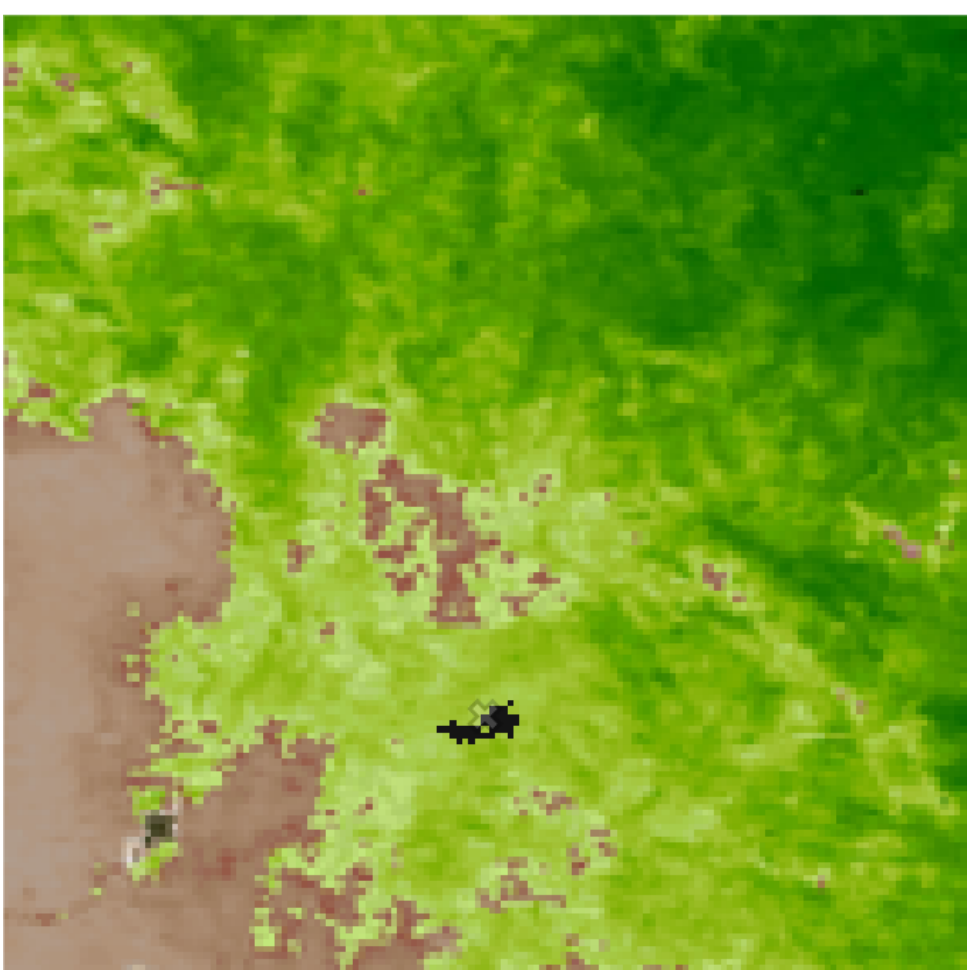
Simulations



Regular climate conditions, high flammability and no wind.

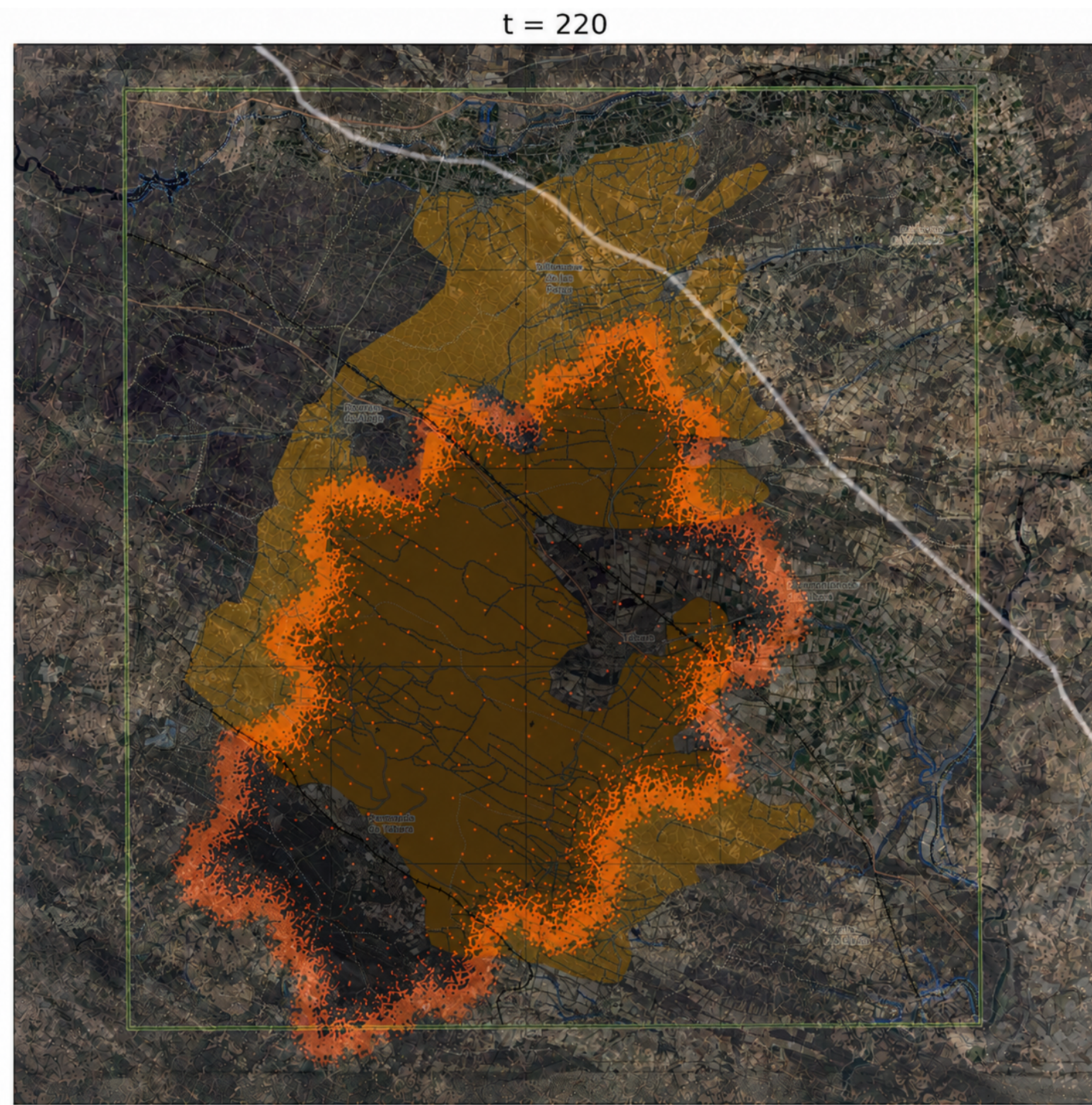


Regular climate conditions, high flammability and wind from the left.



Regular climate conditions, low-flammability ignition point.

Model validity vs. real wildfire spread



Real vs. Model Wildfire Spread — Zamora 2022

The image compares the real wildfire spread, represented by the transparent yellow perimeter, with the wildfire propagation generated by our stochastic spatial SIR model, shown by the black and orange burned region. The comparison with satellite-derived wildfire perimeters shows that the model reproduces the main spread geometry, propagation direction, and overall magnitude of the burned region.

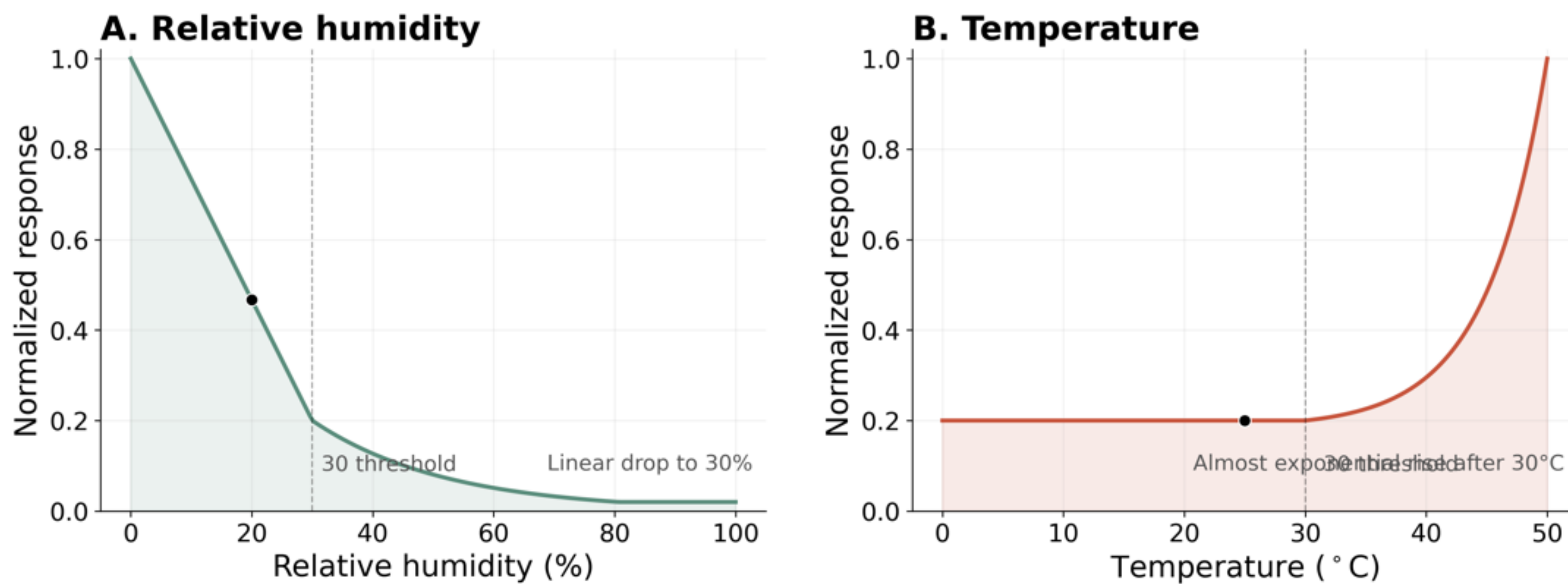
Although simplified, the model captures realistic large-scale propagation patterns through local ignition, neighborhood interactions, spatial flammability, and wind-driven anisotropy. Several real-world mechanisms, such as firefighter intervention, suppression strategies, and rapidly changing atmospheric conditions, are not explicitly included.

Key Findings and Implications

The simulations highlight the importance of terrain heterogeneity and wind-driven propagation under identical California climate conditions:

- Higher local flammability generates significantly larger burned regions over the same number of timesteps
- Wind produces elongated anisotropic propagation patterns by pushing the fire front in a dominant direction
- Ignition in low-flammability regions causes the fire to extinguish rapidly.

Wildfire threshold behavior and climate interaction



The graphs show how the model also reveals strong nonlinear threshold behavior:

- Temperatures above approximately 30°C,
- Relative humidity below approximately 30%,
- Strong directional wind conditions

substantially amplify wildfire propagation and generate persistent fire fronts.

Conclusions

This project demonstrates that stochastic spatial SIR models can reproduce realistic wildfire propagation under heterogeneous environmental conditions.

By combining local stochastic interactions, satellite-derived flammability maps, wind-driven anisotropy, and climate-dependent ignition dynamics, the model captures several qualitative behaviors observed in real wildfires. In particular, the simulations reveal strong threshold behavior associated with high temperatures, low humidity, and intense ignition conditions.

The results show that wildfire propagation is highly sensitive to terrain heterogeneity and environmental forcing. Wind produces anisotropic spread patterns and elongated fire fronts, while temperature and humidity strongly affect ignition persistence and propagation velocity.

Using real satellite imagery preserves vegetation structure across different landscapes, including California, Australia, and Zamora. Although simplified, the model provides a flexible framework for studying wildfire spread through stochastic dynamics and spatial data.

References

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